

Artificial Intelligence: The Technology That's Changing Our World

Hello everyone! Today, we're going to explore something that's both fascinating and increasingly important in our daily lives - Artificial Intelligence, or AI. Think of AI as teaching computers to think and learn, similar to how humans do, but in their own unique way. It's like giving machines a brain that can process information, recognise patterns, and make decisions.

 by Uday Kumar

Use Case

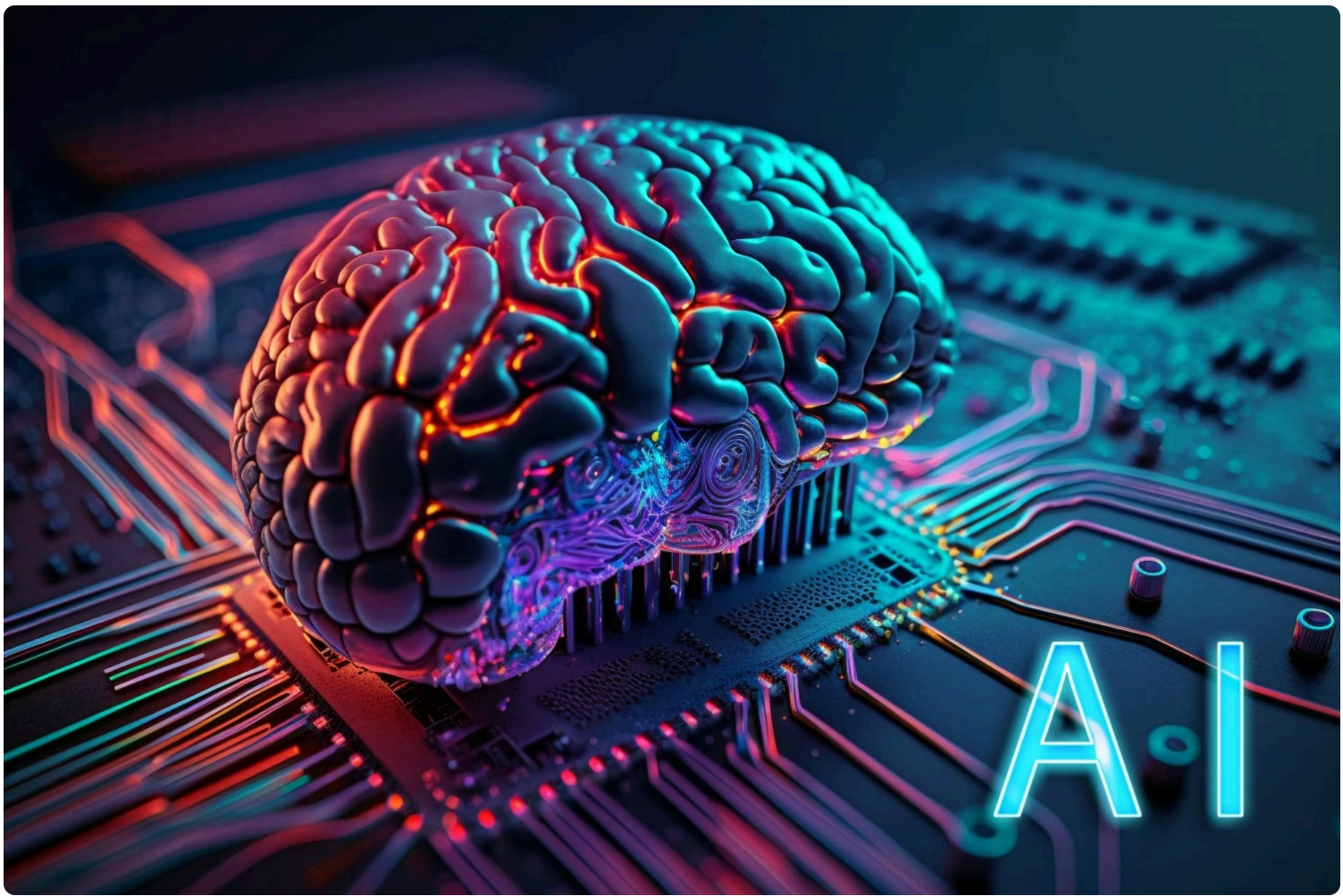
Let me start with something many of you probably used this morning - your smartphone's autocomplete feature. Have you noticed how it seems to know what you're going to type next? That's AI in action! My nephew recently sent a text to his mom saying "On my way," and before he could type "home," his phone suggested exactly that word. This simple example shows how AI learns from our patterns and habits to make our lives easier.

This kind of AI is called machine learning - the computer learns from millions of text messages to predict what words typically come next. It's like having a very attentive friend who remembers everything you've ever said!



How does AI work?

Think of an AI (LLM) like a super-advanced pattern recognition system that has read through millions of books, articles, and websites.



Imagine a student who has read every book in the world's largest library. When you ask this student a question, they don't actually "know" the answer, but they can piece together relevant information from everything they've read.

Training Data

- Billions of text examples from:
 - * Books
 - * Websites
 - * Articles
 - * Social media
 - * Code
 - * Other online content

Analogy: Think of it as feeding the AI a massive digital library.

Pattern Recognition

- Learns which words commonly appear together
 - Understands context and relationships
- Analogy: Like learning that:
- "Hot" often comes with "coffee"
 - "Good" often comes with "morning"

Prediction Process

1. Reads input text
2. Identifies patterns
3. Predicts most likely next words
4. Builds response word by word

Simple Example:

Input: "The sky is..."

Model thinks:

- "blue" (highest probability)
- "cloudy" (second highest)
- "dark" (third highest)

Types AI Tools

1 Text to Text Generation





ChatGPT

A conversational AI system that listens, learns, and challenges





Claude

Talk with Claude, an AI assistant from Anthropic

Supercharge  your content creation with





GravityWrite: Top AI Writer, Content Generator & A...

Use AI to create high-quality content for blogs, ads, emails, and social media in seconds. Boost your clicks, conversion...

2 Text to Image Generation





ChatGPT

A conversational AI system that listens, learns, and challenges



Free AI Image Generator – Turn Text to Image Onlin...

Generate stunning AI images and art from text prompts or images online with Stable Diffusion's AI image generator....







AI Image Generator – Create Art, Images & Video | ...

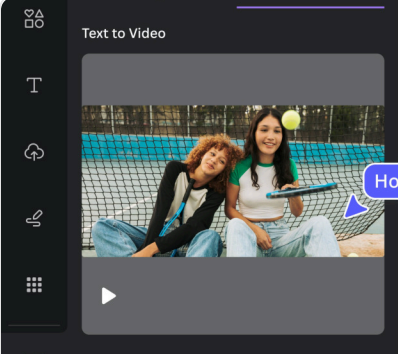
Transform your projects with our AI image generator. Generate high-quality, AI generated images with...

3 Text to Video Generation



Free AI Video Generator | invideo AI

Generate AI videos with just a text using invideo's AI video generator. It generates the script and adds video clips,...





AI Video Generator: Text to Video AI Tool | Canva

Instantly create an AI-generated video from a text prompt with the AI text-to-video generator on Canva.





Sora: Creating video from text

Sora is an AI model that can create realistic and imaginative scenes from text instructions.



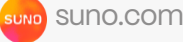


Runway

Everything you need to make anything you want.

4 Text to Audio Generator





Suno

Suno is building a future where anyone can make great music.



Voice Generator (Online & Free) 

Generate voice from text and play or download the resulting audio file. It's all online, and completely free! This...



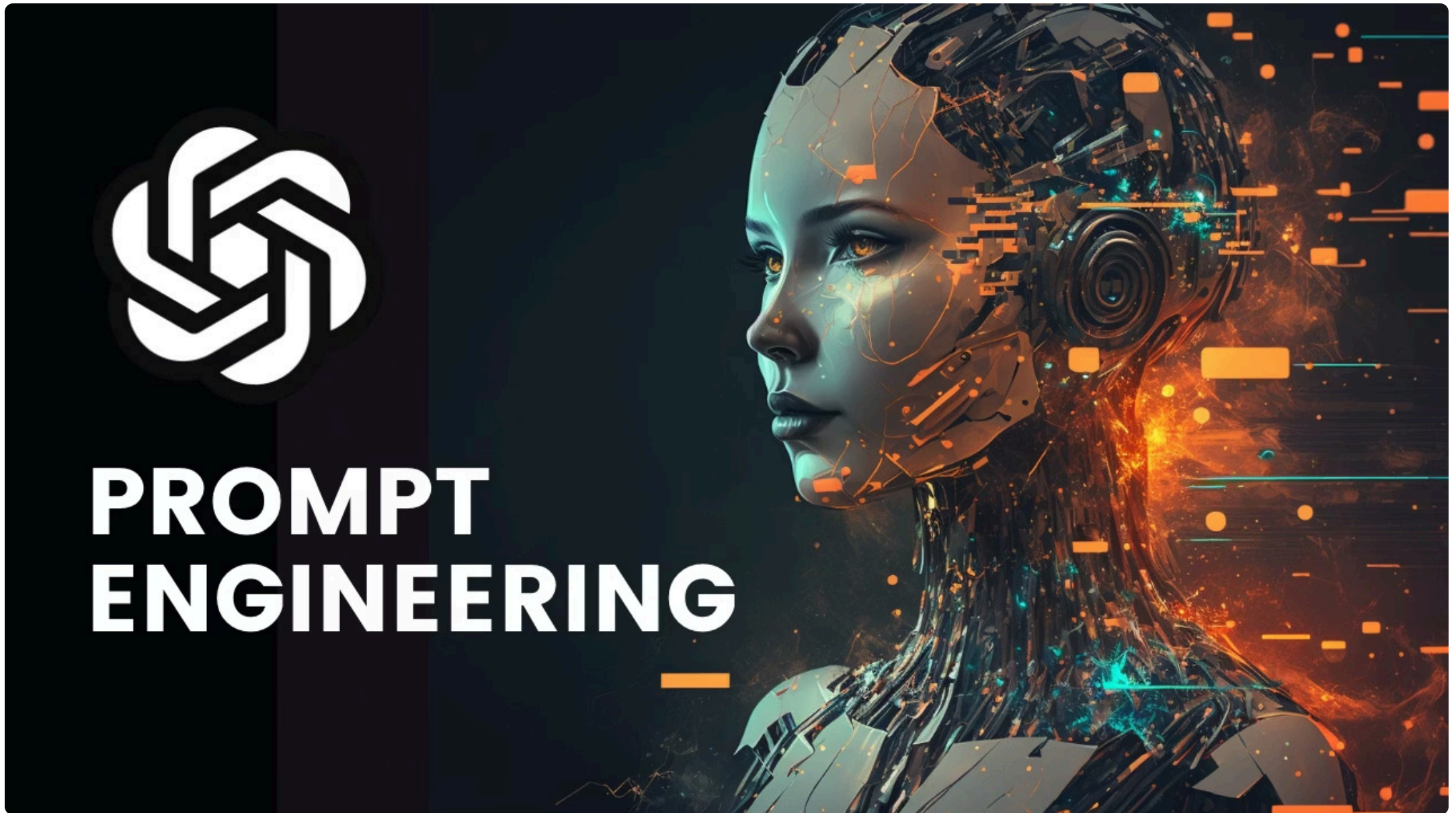


ElevenLabs: Free Text to Speech & AI Voice Genera...

Create the most realistic speech with our AI audio in 1000s of voices and 32 languages. Pioneering research in Text to...

PROMPT ENGINEERING FUNDAMENTALS

Prompt engineering is the skill of crafting effective instructions for AI models to get desired outputs. It's like learning to communicate with AI in the most effective way.



Example: Write about climate change on earth?

PROMPTING TECHNIQUES

1. ROLE PROMPTING



Technique: Assigning a specific role to the AI

Example:

✗ Basic: "Tell me about investment strategies."

✓ Better: "Act as a certified financial advisor with 20 years of experience. Explain investment strategies for a beginner with \$5000 to invest."

2. CHAIN-OF-THOUGHT PROMPTING



Technique: Breaking down complex problems into steps

Example:

✗ Basic: "Solve this math problem: If a store offers 30% discount and then adds 20% tax, what's the final price of a \$100 item?"

✓ Better: "Let's solve this step by step:

1. Start with the original price: \$100
2. Calculate 30% discount
3. Subtract discount from original price
4. Calculate 20% tax on the discounted price
5. Add tax to get final price"

3. FEW-SHOT PROMPTING



Technique: Providing examples of desired output

Example:

✗ Basic: "Write product descriptions."

✓ Better:

"Write product descriptions following these examples:

Example 1:

Product: Coffee Mug

Description: Sleek ceramic mug in midnight blue, holds 12 oz of your favorite beverage. Microwave-safe with ergonomic handle.

Example 2:

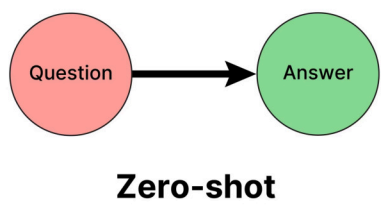
Product: Yoga Mat

Description: Premium non-slip yoga mat in sage green, 6mm thick for extra comfort. Includes carrying strap.

Now write a description for:

Product: Wireless Headphones"

4. ZERO-SHOT PROMPTING



Technique: Getting results without examples

Example:

- ✗

Basic: "Write about climate change."
- ✓

Better: "Explain the causes and effects of climate change in simple terms that a 12-year-old would understand. Include three main points and use relevant analogies."

5. FORMAT SPECIFICATION

LAYERED FILES

Common formats for saving your work in Photoshop.

PSD

PSB

TIFF

STANDARD WEB FILES

Common formats for displaying images on the web.

JPG

PNG

GIF

Technique: Clearly defining output format

Example:

- ✗

Basic: "Give me information about healthy foods."
- ✓

Better: "Create a table of 5 superfoods with the following columns:

- Food Name

- Key Nutrients

- Daily Recommended Portion

- Health Benefits

- Easy Recipe Suggestion"

6. CONTEXT SETTING

Technique: Providing background information

Example:

✗ Basic: "How can I improve my presentation?"

✓ Better: "I'm giving a 15-minute technical presentation to non-technical executives about cybersecurity. My goal is to secure budget for new security measures. How can I make my presentation more effective?"

7. PERSONA-BASED PROMPTING



Technique: Tailoring content for specific audiences

Example:

✗ Basic: "Explain how a car engine works."

✓ Better: "Explain how a car engine works to a group of curious 10-year-olds who love cars but have no technical knowledge. Use simple analogies and avoid technical jargon."

8. TEMPLATE PROMPTING

Technique: Using consistent structures

Example:

...

Template:

Topic: [Insert Topic]

Target Audience: [Insert Audience]

Tone: [Insert Tone]

Length: [Insert Length]

Special Instructions: [Insert Any Special Requirements]

Required Sections:

1. Introduction

2. Main Points

3. Examples

4. Conclusion

...

9. CONSTRAINT PROMPTING



Technique: Setting specific limitations

Example:

✗ Basic: "Write a story about adventure."

✓ Better: "Write a 300-word adventure story with:

- No violence
- Suitable for children under 10
- Must include a moral lesson
- Set in a forest
- Maximum 3 characters"

10. OUTPUT MODIFICATION

Technique: Refining generated content

Example:

✗ Basic: "Make this more interesting."

✓ Better: "Revise the following text to:

1. Use more vivid adjectives
2. Include sensory details
3. Vary sentence structure
4. Add two relevant metaphors
5. Maintain a professional tone"

PROMPT FRAMEWORK

R-T-F → Act as a (ROLE) Create a (TASK) Show as (Format)

Prompt example:

Act as Facebook Ad Marketer

Design a compelling Facebook ad campaign to promote

A new line of fitness apparel for a sports brand

Create a storyboard that outlines the sequence of ad creatives

Including ad copy, visuals, and a targeting strategy.

T-A-G → Define a (TASK) State the (ACTION)

Prompt example

The task is to evaluate the performance of team members.

Act as a direct manager and assess the strengths and weaknesses of the team members

Goal is to improve team performance so that the average

user satisfaction score moves from 6 to 7.5 in the next quarter

B-A-B → Explain problem (BEFORE State outcome (AFTER)

Prompt example

We're nowhere to be seen in SEO rankings

We want to be in the top 10 SEO rankings in our niche within 90 days

Develop a detailed plan for mentioning all the measures.

we should also include a list of the top 20 keyboards.

C-A-R-E → Give the (CONTEXT) Describe (ACTION) Clarify the (RESULT) Give the (EXAMPLE)

Prompt example

We are launching a new line of sustainable clothing

Can you assist us in creating a targeted advertising

campaign that emphasizes our environmental commitment?

Our desired outcome is to drive product awareness and sales A good example of a similar successful initiative is Patagonia's "Don't Buy This Jacket" campaign, which highlighted their commitment to sustainability while enhancing their brand image.

ChatGPT Prompting Cheat Sheet

1. Tone: Specify the desired tone (e.g., formal, casual, informative, persuasive).
2. Format: Define the format or structure (e.g., essay, bullet points, outline).
3. Act as: Indicate a role or perspective to adopt (e.g., expert, critic, enthusiast).
4. Objective: State the goal or purpose of the response (e.g., inform, persuade).
5. Context: Provide background information, data, or context for content generation.
6. Scope: Define the scope or range of the topic.
7. Keywords: List important keywords or phrases to be included.
8. Limitations: Specify constraints, such as word or character count.
9. Examples: Provide examples of desired style, structure, or content.
10. Deadline: Mention deadlines or time frames for time-sensitive responses.
11. Audience: Specify the target audience for tailored content.
12. Language: Indicate the language for the response, if different from the prompt.
13. Citations: Request the inclusion of citations or sources to support information.
14. Points of view: Ask AI to consider multiple perspectives or opinions.
15. Counterarguments: Request addressing potential counterarguments.
16. Terminology: Specify industry-specific or technical terms to use or avoid.
17. Analogies: Ask AI to use analogies or examples to clarify concepts.
18. Quotes: Request inclusion of relevant quotes or statements from experts.
19. Statistics: Encourage the use of statistics or data to support claims.
20. Call to action: Request a clear call to action or next steps.
21. Questions: Have the AI ask you questions for further clarification or direction

ChatGPT Prompting Cheat Sheet

EXPLAIN LIKE I'M 5



Description: Break down tough ideas so even a kid could get it. Perfect for when you're totally lost on a topic.

Prompt Template: "Explain [insert concept or topic] as if you were talking to a 5-year-old child. Use simple language and everyday examples."

ROLE-PLAY



Description: Act out scenarios to practice what you've learned.

Prompt Template: "Let's role-play a scenario where I'm [insert role] and you're [insert another role] We'll practice [insert skill or situation]. Begin the scenario, and i'll respond accordingly."

EXAMPLES AND ANALOGIES



Description: Get real-world examples that make abstract ideas click.

Prompt Template: *Explain (insert concept or topic) using three different real-world examples or analogies that would be easy for a beginner to understand."

MOTIVATION



Description: Get tips to keep yourself motivated about learning, even when it gets tough

Prompt Template: "I'm struggling to stay motivated while learning [insert subject or skill]. Provide me with 5 practical strategies to boost my motivation and maintain consistency in my studies."

STUDY PLAN



Description: Map out your learning journey from start to finish Great for tackling big subjects without getting overwhelmed

Prompt Template: "Create a detailed study plan for learning [insert subject or skill] over the course of [insert time frame]. Include specific goals, resources, and milestones."

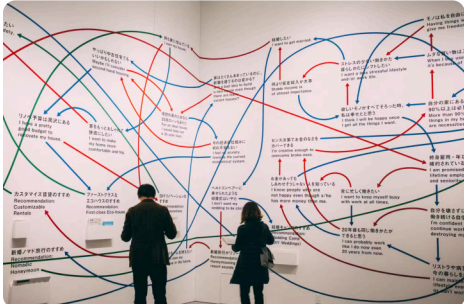
Quiz



Description: Create some practice questions to see if you've really got the hang of things.

Prompt Template: *Generate a 10-question quiz on [insert topic], including a mix of multiple-choice, true/false, and short-answer questions. Provide answers and brief explanations for each question."

MINDMAP



Description: Sketch out your thoughts to see how everything connects. Really helpful when you're trying to wrap your head around a big topic.

Prompt Template: *Create a detailed mind map for the topic [insert topic] Include main branches, sub-branches, and key concepts or ideas for each."

EXPERT ROUNDTABLE



Description: Simulate a discussion between people who really know their stuff. Great for getting different views on tricky topics.

Prompt Template: "Simulate a roundtable discussion with me and three experts in finsert field] discussing [insert topic or question]. Present their different viewpoints and any potential areas of agreement or disagreement."

MENTAL ASSOCIATIONS



Description: Come up with clever memory tricks to help stuff stick in your brain. Useful for remembering lists, facts, or tricky concepts.

Prompt Template: "Help me create mental associations or mnemonic devices to remember key information about [insert topic or concept]."

IMPROVE WHAT YOU HAVE



Description: Get feedback and suggestions to make your work even better. Great for polishing up something you've already created.

Prompt Template: "Here's something I've (written/created/produced]: [insert your work]. Please provide specific suggestions to improve it, focusing on [aspect you want to improve, e.g., clarity, structure, persuasiveness). Explain why each chang would make it better."

ChatGPT Prompting Cheat Sheet

Market Research

"Can you conduct a detailed analysis of the latest trends in [specific market or industry, highlighting the top 5 emerging opportunities for new entrepreneurs looking to enter the market?"

Content Creation

"Could you assist in drafting a comprehensive blog post on [specific topic], ensuring it's engaging and SEO-optimized to significantly increase organic traffic to my website?"

Investment Analysis

"Can you provide a detailed analysis of [specific investment opportunity or market], including potential risks, expected returns, and recommendations for a novice investor?"

Online Course

"Given my expertise in [user's field], what specific topics would be most lucrative for developing online courses, considering current market demand and competition?"

Idea Generation

"Considering the current market trends, what are some unique and creative business ideas in [user's field of interest] that offer high profitability with minimal competition?"

Social Media Strategy

*What specific strategies should I employ on [specific social media platform] to enhance my [business or personal brand's] presence, increase follower engagement, and drive conversions?"

Freelancing

"In the realm of (user's skill set), what are the current high-paying freelancing opportunities, and how can I effectively market myself to attract premium clients and projects?"

Copywriting for sales

"How can I craft a persuasive and compelling sales copy for [user's product or service] that effectively communicates value and drives high conversion rates among potential customers?"

Product Development

"What innovative features or improvements can be added to [existing product or service] to better meet current market demands and distinguish it from competitors in a crowded marketplace?"

E-commerce Store

"How can I strategically optimize my e-commerce store, focusing on (specific product category). to improve user experience, increase conversion rates, and boost overall sales?"

Affiliate Marketing

"What are the most effective strategies for starting an affiliate marketing venture in the niche of (user's area of interest), and which platforms offer the best partnership opportunities?"

Strategic Networking

"What strategies can I employ to effectively network within [user's industry], seeking out valuable partnerships and collaborative opportunities to expand my business reach and impact?"